

Roshni Singh

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EXPERIENCE

CrashPlan - Junior Software Engineer
- Developed an API to retrieve and filter Kibana/Elasticsearch logs by error type, environment, agent, batch ID, transaction ID, and other operational attributes, providing a structured interface for investigating error data. - Designed the log-storage workflow to persist searchable error metadata in MongoDB while storing complete log payloads in Azure Blob Storage, organised around batch and transaction identifiers. - Built a crawler to process large volumes of Elasticsearch logs, implementing pagination to overcome the 10,000-result retrieval limit and ensure complete log extraction for transactions containing large numbers of logs. Added persistent checkpoints in a dedicated MongoDB collection to allow interrupted crawls to resume from their last successful position. - Implemented error-lifecycle logic to determine whether errors are active or resolved, ensuring only relevant ongoing incidents are surfaced for further processing. - Developed a prompter workflow for Lucy, the company's MCP-based AI assistant, to send structured context about active errors and persist Lucy's responses, including error details, reasons for continued activity, and suggested resolution actions. - Developed a role-based authorisation layer for Lucy to validate user permissions before allowing access to potentially restricted operational information.

Mathological Technologies Private Limited - Engineering Intern
- Worked on an AI-driven predictive maintenance system for Metro Rail assets using fault-event and IoT sensor data. - Performed EDA, data validation, visualization, preprocessing, data cleaning, and feature engineering to prepare sensor and maintenance data for ML models. - Researched predictive maintenance, failure prediction, and machine learning methodologies for rail asset monitoring. - Collaborated with the team to support the development and analysis of predictive maintenance solutions.

EDUCATION

Vellore Institute of Technology B.Tech – Computer Science and Engineering	2022 – 2026
Delhi Public School Bangalore South Senior Secondary Education (12th)	2022

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL
Frontend/3D rendering: React.js, Next.js, Three.js
Backend & Real-Time: Node.js, Express.js, FastAPI, REST APIs, WebSockets, MCP
Databases & Storage: PostgreSQL, MongoDB, Redis, Azure Blob Storage, Elasticsearch, ChromaDB, Qdrant
Cloud & DevOps: Microsoft Azure, AWS, Docker, CI/CD
AI / ML: PyTorch, CNNs, Transformers, Deep Learning, Reinforcement Learning, PSD Analysis, Signal Processing
GenAI: RAG, Agentic RAG, LangChain, LangGraph, HuggingFace, Ollama, LLM APIs, Prompt Engineering, Vector Embeddings
Architecture & Systems: Client-Server Architecture, Server-Authoritative Architecture, Real-Time State Synchronization, Distributed Systems, Model-Agnostic LLM Architecture
Security: Authentication, Authorization, Role-Based Access Control (RBAC), JWT
Tools: Git, GitHub, Kibana, SUMO, Unity, Postman

PROJECTS

DevClash – Multiplayer Competitive Engineering Platform
Stack: React.js, Node.js, Express.js, TypeScript, MongoDB, Redis, WebSockets, JWT
Building a competitive software-engineering platform combining DSA, coding, debugging, SQL, output prediction, and realistic engineering challenges across solo, ranked multiplayer, private-room, and team-based modes.

- ▶ Designing a modular challenge architecture where questions and engineering scenarios are treated as data rather than application logic, with a separate DevClash-Bank for challenge content, test cases, logs, code, database states, and evaluation metadata.
- ▶ Developing an Admin Panel and challenge lifecycle for creating, reviewing, publishing, versioning, archiving, and analyzing challenges without requiring changes to the core application codebase.
- ▶ Building real-time competitive gameplay using WebSockets, with matchmaking, private rooms, synchronized timers, live match state, player presence, reconnect handling, scoring, and rating-based progression.
- ▶ Designing Redis-backed real-time infrastructure for matchmaking queues, temporary game state, presence, caching, pub/sub, and scalable WebSocket communication alongside MongoDB-backed persistent application data.
- ▶ Developing Team Mode around realistic production incidents involving API failures, authentication/authorization issues, database problems, security vulnerabilities, performance bottlenecks, and deployment failures, requiring teams to investigate code, logs, API responses, configuration, and system state.
- ▶ Designing a secure submission, evaluation, and progression pipeline for coding and debugging challenges using isolated execution, automated tests, resource limits, scoring, MMR, leagues, achievements, leaderboards, and persistent match history.

Last Stand – Real-Time Multiplayer Social Strategy Game [GitHub ↗](#)

Stack: Three.js, Redis, TypeScript, React.js, Node.js, WebSockets, PostgreSQL

- ▶ Built a real-time multiplayer web platform using Three.js and React for the interactive client, with a TypeScript/Node.js backend responsible for authoritative game state and multiplayer coordination.
- ▶ Designed a server-authoritative architecture where clients submit player actions, the backend validates inputs and applies state transitions, and synchronised updates are propagated to connected players through WebSockets.
- ▶ Developed reusable real-time services for matchmaking, private rooms, lobbies, persistent sessions, player presence, reconnection, and match lifecycle management, enabling multiple multiplayer experiences to share the same infrastructure.
- ▶ Integrated Redis for ephemeral multiplayer state, matchmaking, presence, caching, and pub/sub, while using PostgreSQL for persistent player accounts, match history, progression, and telemetry.
- ▶ Defined versioned, schema-first WebSocket contracts between the web client and backend to maintain consistent real-time communication and predictable state synchronization.
- ▶ Developed Island, a multiplayer social-strategy experience featuring hidden roles, cooperative tasks, fragmented information, real-time communication, constrained voting, tournament progression, and AI-controlled opponents.
- ▶ Designed the platform around a reusable input → validation → state transition → synchronization pipeline, allowing gameplay mechanics to evolve without tightly coupling client behavior to game rules.

InsightPDF – Agentic RAG Document Intelligence Platform [GitHub ↗](#)

Stack: Next.js, TypeScript, Python, LangChain, LangGraph, ChromaDB, HuggingFace, Ollama, LLM APIs, PostgreSQL

- ▶ Built an agentic RAG platform for large document collections, combining retrieval, extraction, analysis, comparison, research, and evidence-backed generation through stateful LangGraph workflows.
- ▶ Developed an agentic tool layer for semantic retrieval, page/section lookup, structured extraction, calculations, document comparison, web research, and report generation, allowing workflows to select tools dynamically based on user intent.
- ▶ Built document-aware learning workflows generating summaries, notes, quizzes, flashcards, and question papers, with outputs grounded in the selected document or chapter.
- ▶ Implemented iterative retrieval and validation that evaluates evidence quality, refines queries when necessary, and produces document- and page-level citations to reduce unsupported responses.
- ▶ Developed a model-agnostic LLM gateway supporting cloud APIs with automatic fallback to local Ollama models, reducing dependency on a single model provider and improving availability.
- ▶ Separated deterministic processing such as document parsing, retrieval, validation, calculations, and structured outputs from agentic planning and multi-step execution for greater control over workflow behavior.
- ▶ Built an evaluation and observability layer measuring retrieval relevance, answer faithfulness, citation accuracy, latency, model usage, and agent/tool execution.

ACHIEVEMENTS & CERTIFICATIONS

IBM Generative AI Professional Certificate

Solved 500+ problems on [LeetCode](#) & [GeeksforGeeks](#)